

RESEARCH ARTICLE

Quaternion-based adaptive control for trajectory tracking of quadrotor unmanned aerial vehicles

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Summary

The problem of trajectory tracking of aerial vehicles with four rotors and dynamic parameter uncertainties is addressed in this article. By exploiting the cascade structure of the translational and attitude dynamics of the quadrotor, a nonlinear hierarchical adaptive control is proposed. The attitude control law is designed based on the unit quaternion, therefore, the singularities of the Euler angles and angle-axis representation are avoided. Instead of relying on the stability properties of time-varying cascade systems, the stability of the whole closed-loop system is proven by an appropriate Lyapunov function. In addition, a novel method to compute the desired orientation based on the desired Cartesian trajectories and position control inputs is proposed. A set of simulations results are presented to assess the performance of the proposed adaptive controller.

KEYWORDS

adaptive control, quadrotor, trajectory tracking, unit quaternion

1 | INTRODUCTION

The quadrotor is an unmanned aerial vehicle (UAV) that can perform a variety of complex maneuvers in a tridimensional environment such as vertical take-off and landing, hovering, and aggressive flights among others. For these reasons, the quadrotor has become very popular in recent years and it has been used in several applications: object transportation, surveillance and recognition tasks, rescue missions, filming just for mention a few. The differential equations that describes the quadrotor's motion are highly nonlinear.¹ Moreover, the system is underactuated with only four inputs and six degrees of freedom. As result, the trajectory control problem of this vehicles is a challenging task.

In recent years, several control laws for trajectory tracking and stabilization of this type of aircraft have been proposed in the literature. PID and LQR controllers have been demonstrated to be effective for stabilization tasks and in certain hovering conditions.²⁻⁴ In order to perform complex maneuvers, nonlinear control techniques have been employed. Nonlinear feedback linearization schemes have been commonly used for trajectory tracking.^{5,6} The quadrotor's dynamics presents a cascade structure between its translational and attitude dynamics. This property has been exploited to design backstepping controllers⁷⁻⁹ and inner/outer-loops control schemes.^{10,11} The aforementioned controllers are designed based on the Euler angles. Such parametrization of the orientation suffers from singularities. To overcome this problem, Lee et al.¹² propose a geometric tracking controller on the special Euclidean group $SE(3)$ which is almost globally attractive. The attitude errors are defined in terms of the rotation matrix. Similar control approaches are proposed in the works.^{13,14} Alternatively to the Euler angle and the rotation matrix, Roberts and Tayebi¹⁵ propose an adaptive nonlinear controller base on the unit quaternion parametrization where the adaptive term estimates bounded constant disturbances.

The control laws mentioned so far rely on the exact knowledge of the aircraft dynamic parameters. Nevertheless, in practice there are always model uncertainties. One possible solution to address parameter uncertainties and unmodeled dynamics is to design robust controllers.^{16,17} Sliding mode control theory is often used to design robust controllers for trajectory tracking of quadrotors.^{18,19} However, most of them adopt the Euler angles parametrization. On the other hand, adaptive controllers can be used to address the problem of parametric uncertainty. In this regard, MRAC controllers have been developed for trajectory tracking of UAVs.²⁰⁻²² A fuzzy adaptive backstepping control algorithm was proposed in the work.²³ The proposed controller compensates the mass uncertainty and rejects time-varying disturbances. Perez and Moreno²⁴ designed a nonlinear hierarchical adaptive control that ensures trajectory tracking. The developed theory was evaluated through a set of experiments. Nevertheless, due to the singularities of the Euler angles these adaptive controllers cannot be used for aggressive maneuvers. An adaptive control based on the unit quaternion is proposed in the work.²⁵ The controller exploits the cascade structure of the quadrotor dynamics. The stability analysis is based on the stability properties of time-varying cascade systems. However, the equilibrium points of translational and attitude subsystems are only stable in the sense of Lyapunov. Therefore, the results given by Loria and Panteley²⁶ cannot be directly applied. Ha et al.²⁷ propose a passivity based tracking control for quadrotor UAVs with unknown mass, but only the attitude kinematics were considered. This control law is suitable for commercial quadrotors where the control inputs are the angular velocities. Song et al.²⁸ propose a novel passivity based adaptive control for quadrotor with mass and inertia uncertainties. A robust nonlinear control law with an integral sliding mode is proposed by Ibarra-Jimenez et al..²⁹ The main advantages of the controller are that the trajectory tracking is achieved in finite time and is robust against external disturbances. To cope with the problem of system uncertainties in quadrotor aircrafts, Jin et al.³⁰ propose an adaptive controller based on neural network (NN) techniques. An adaptive PI controller in combination with a self-tuning regulator (STR) is proposed by Lei et al..³¹ Due to its structure, the STR controller is easy to implement in a microcontroller or an embedded system. Nevertheless, the aforementioned control algorithms suffer from the singularities of the Euler angles representation and the results are only local. Recently, an adaptive control law based on the unit quaternion which does not require linear velocity measurements is presented by Hirata-Acosta et al..³² However, the authors consider only uncertainties in the inertia matrix of the aerial vehicle. Finally, a survey on nonlinear and adaptive controllers for UAVs can be found in the work.³³

The problem of trajectory tracking of quadrotors UAVs with mass and inertia matrix uncertainties is addressed in this article. Due to the cascade structure of the translational and attitude dynamics of the aerial vehicle a hierarchical control strategy was adopted. For the translational dynamics, a virtual control input is introduced to overcome the under-actuated nature of the system. On the other hand, the attitude controller is based on the unit quaternion and takes into account not only the attitude tracking errors but also the tracking errors of the translational subsystem. This is a significant difference over inner-outer control laws proposed in the literature where the position and attitude controllers are designed separately. To cope with the model uncertainties two Lyapunov-based adaptive laws are proposed. Contrary to other works where the stability proof is based on the stability properties of time-varying cascade systems, in this work, an appropriate Lyapunov function is proposed for the whole closed-loop system. The stability analysis shows that the closed-loop trajectories are bounded and the position and attitude errors converge asymptotically to zero. In summary, the main contributions of the article are the following: 1) the proposed control laws solve the problem of trajectory tracking for quadrotors with model uncertainties. 2) The attitude controller is based on the unit quaternion, as a result, the controller is singularity free and it allows to perform complex maneuvers. 3) An alternative method is proposed to compute the desired orientation of the vehicle which gives not only the desired direction of the total thrust force necessarily to track the desired Cartesian trajectory but also provides an additional degree of freedom which plays a similar role of the yaw angle.

The remainder of the article is organized as follows: the dynamic model and properties of the quadrotor is presented in Section 2. The proposed adaptive controller with its corresponding stability analysis is developed in Section 3. In Section 4 simulation results are presented to validate the proposed theory. Finally some concluding remarks and future directions are presented in Section 5.

2 | EQUATIONS OF MOTION

In this section, the dynamic model of a four-rotor aerial vehicle is presented. The quadrotor can be modeled as rigid body in a three dimensional space. Let $\mathcal{I} = \{\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i\}$ be an inertial frame and let $\mathcal{B} = \{\mathbf{x}_b, \mathbf{y}_b, \mathbf{z}_b\}$ be a second coordinate frame attached to the quadrotor's center of mass (see Figure 1). The coordinates frames $\mathcal{I}$ and $\mathcal{B}$ are related by the rotation matrix

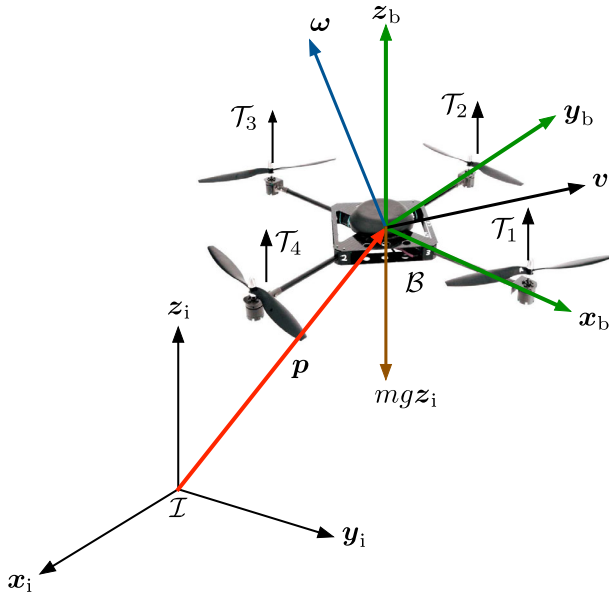


FIGURE 1 Parameters and state variables of the quadrotor
[Colour figure can be viewed at wileyonlinelibrary.com]

$R \in SO(3) = \{R \in \mathbb{R}^{3 \times 3} | R^T R = I, \det(R) = 1\}$ which describes the orientation of the quadrotor. The position and linear velocity of the quadrotor's center of mass are described by the vectors $p \in \mathbb{R}^3$ and $v \in \mathbb{R}^3$, respectively. Both vectors are expressed with respect to the inertial frame I . In Figure 1, the vector $\omega \in \mathbb{R}^3$ represents the angular velocity expressed with respect to the body frame B . The translational equations of motion of the vehicle are given by

$$\dot{p} = v, \quad (1a)$$

$$m\dot{v} = \tau R z_i - mg z_i, \quad (1b)$$

where g and m are the gravity constant and the mass of the quadrotor, respectively. The control input for the translational dynamics is the total thrust $\tau = \sum_{i=1}^4 \tau_i$ generated by the rotors. The attitude equations of motion expressed with respect to the body frame are the following:

$$\dot{R} = RS(\omega), \quad (2a)$$

$$J\dot{\omega} = \tau - S(\omega)J\omega, \quad (2b)$$

where $J \in \mathbb{R}^{3 \times 3}$ is the inertia matrix and $\tau \in \mathbb{R}^3$ is the torque generated by rotors. Finally, the operator $S(\cdot) \in \mathfrak{so}(3) = \{S \in \mathbb{R}^{3 \times 3} | S + S^T = O\}$ is defined as

$$S(\vartheta) = \begin{bmatrix} 0 & -\vartheta_z & \vartheta_y \\ \vartheta_z & 0 & -\vartheta_x \\ -\vartheta_y & \vartheta_x & 0 \end{bmatrix}, \quad \forall \vartheta = \text{col}(\vartheta_x, \vartheta_y, \vartheta_z) \in \mathbb{R}^3$$

and satisfies $S(\vartheta_1)\vartheta_2 = \vartheta_1 \times \vartheta_2$ and $\|S(\vartheta_1)\| = \|\vartheta_1\|$ for all $\vartheta_1, \vartheta_2 \in \mathbb{R}^3$. It is important to point out that the subsystem (1) is underactuated with only one control input τ , while the attitude subsystem (2) is completely actuated with input τ . The term $\tau R z_i$ is the coupling or interconnection term between each subsystem and gives the magnitude and direction of the total force generated by the rotors. Finally, the subsystem (2) presents the following property:

Property 1. The attitude dynamics described by (2) is linear with respect to the inertia parameters, that is,

$$J\dot{\omega} + S(\omega)J\omega = \Psi_o(\omega, \dot{\omega})\theta = \tau, \quad (3)$$

where $\theta \in \mathbb{R}^p$ is the constant inertia parameter vector and $\Psi_o(\omega, \dot{\omega}) \in \mathbb{R}^{3 \times p}$ is called the regressor matrix.

Alternatively to the rotation matrix $\mathbf{R}$, the orientation of the vehicle can be described by a four-parameter representation called the unit quaternion denoted by $\mathcal{Q} = \{\eta, \epsilon\} \in \mathbb{S}^3$, where $\eta \in \mathbb{R}$ and $\epsilon = \text{col}(\epsilon_x, \epsilon_y, \epsilon_z) \in \mathbb{R}^3$ are the scalar and vector parts, respectively¹. Given the unit quaternions $\mathcal{Q}_1 = \{\eta_1, \epsilon_1\}$ and $\mathcal{Q}_2 = \{\eta_2, \epsilon_2\}$, the quaternion product is defined as $\mathcal{Q}_1 \otimes \mathcal{Q}_2 = \{\eta_1\eta_2 - \epsilon_1^T \epsilon_2, \eta_1\epsilon_2 + \eta_2\epsilon_1 + \mathbf{S}(\epsilon_1)\epsilon_2\}$. The inverse of $\mathcal{Q}$ denoted by $\mathcal{Q}^{-1}$ is defined as $\mathcal{Q}^{-1} = \{\eta, -\epsilon\}$ with $\mathbf{1}_{\mathcal{Q}} = \{1, \mathbf{0}\}$ as the identity element. Quaternion multiplication is equivalent to matrix multiplication in $SO(3)$. The rotation matrix corresponding to a given quaternion is given by

$$\mathbf{R}(\eta, \epsilon) = (\eta^2 - \epsilon^T \epsilon) \mathbf{I} + 2\epsilon \epsilon^T + 2\eta \mathbf{S}(\epsilon). \quad (4)$$

Let $\mathcal{Q} = \{\eta, \epsilon\} \in \mathbb{S}^3$ and $\boldsymbol{\vartheta} \in \mathbb{R}^3$ the operator $\odot : \mathbb{S}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as $\mathcal{Q} \odot \boldsymbol{\vartheta} = \mathcal{Q} \otimes \{0, \boldsymbol{\vartheta}\} \otimes \mathcal{Q}^{-1} = \mathbf{R}(\eta, \epsilon) \boldsymbol{\vartheta}$ rotates the vector $\boldsymbol{\vartheta}$ about ϵ . Finally, the kinematic relationship between the angular velocity and the time derivative of the unit quaternion are given by

$$\dot{\mathcal{Q}} = \{\dot{\eta}, \dot{\epsilon}\} = \frac{1}{2} \mathcal{Q} \otimes \{0, \boldsymbol{\omega}\} = \mathbf{L}(\eta, \epsilon) \boldsymbol{\omega}, \quad \boldsymbol{\omega} = \mathbf{L}^T(\eta, \epsilon) \begin{bmatrix} \dot{\eta} \\ \dot{\epsilon} \end{bmatrix}, \quad (5)$$

where

$$\mathbf{L}(\eta, \epsilon) \triangleq \frac{1}{2} \begin{bmatrix} -\epsilon^T \\ \eta \mathbf{I} + \mathbf{S}(\epsilon) \end{bmatrix}. \quad (6)$$

3 | ADAPTIVE CONTROLLER

As it was mentioned earlier, the equations of motion of the quadrotor presents a cascade structure with $\mathcal{T} \mathbf{R} \mathbf{z}_i$ as an interconnection term. Moreover, the attitude dynamics (2) is independent of the states $\mathbf{p}$ and $\mathbf{v}$. Such characteristic allows us to design a hierarchical control strategy. The proposed adaptive control law can be separated into an outer-loop (position controller) and an inner-control loop (attitude controller).

3.1 | Position adaptive controller

First, the position and velocity errors are defined as

$$\tilde{\mathbf{p}} = \mathbf{p}_d(t) - \mathbf{p}, \quad \tilde{\mathbf{v}} = \mathbf{v}_d(t) - \mathbf{v} \quad (7)$$

where $\mathbf{p}_d(t) \in \mathbb{R}^3$ and $\mathbf{v}_d(t) = \dot{\mathbf{p}}_d(t) \in \mathbb{R}^3$ are, respectively, the desired bounded position trajectory and desired bounded velocity. Additionally, for control purposes, we define the following reference linear velocity $\mathbf{v}_r$ and auxiliary state $\mathbf{s}_p$ as follows

$$\mathbf{v}_r = \mathbf{v}_d(t) + k_p \tilde{\mathbf{p}}, \quad (8)$$

$$\mathbf{s}_p = \mathbf{v}_r - \mathbf{v} = \tilde{\mathbf{v}} + k_p \tilde{\mathbf{p}}, \quad (9)$$

where k_p is a positive gain. The position subsystem is underactuated, to overcome this problem the following virtual input is introduced

$$\mathbf{f} = \mathcal{T} \mathcal{Q}_d \odot \mathbf{z}_i = \mathcal{T} \mathbf{R}_d(\eta_d, \epsilon_d) \mathbf{z}_i, \quad (10)$$

where $\mathcal{Q}_d \in \mathbb{S}^3$ and $\mathbf{R}_d \in SO(3)$ are the desired unit quaternion and desired rotation matrix, respectively. By taking into account Equations (7)–(10), the translation dynamics (1) can be written as

¹ $\mathbb{S}^n = \{\mathbf{x} \in \mathbb{R}^{n+1} | \mathbf{x}^T \mathbf{x} = 1\}$ and $\text{col} = (\vartheta_1, \vartheta_2, \dots, \vartheta_n)$ denotes a column vector of dimension n .

$$\tilde{\mathbf{v}} = -k_p \tilde{\mathbf{p}} + \mathbf{s}_p, \quad (11a)$$

$$m\dot{\mathbf{s}}_p = m\boldsymbol{\psi}_p(\dot{\mathbf{v}}_r) - \mathbf{f} - \mathcal{T}(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i, \quad (11b)$$

where $\boldsymbol{\psi}_p(\dot{\mathbf{v}}_r) = \dot{\mathbf{v}}_r + g\mathbf{z}_i$ is the translational regressor and $\dot{\mathbf{v}}_r = \dot{\mathbf{v}}_d(t) + k_p \tilde{\mathbf{v}}$. The previous equation can be analyzed as a completely actuated system with input $\mathbf{f}$ perturbed by $\mathcal{T}(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i$. Now, the control objective is to design the virtual input $\mathbf{f}$ such that

$$\lim_{t \rightarrow \infty} \mathbf{s}_p = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\mathbf{p}} = \mathbf{0}.$$

Since the mass of the aerial vehicle is unknown, the following position adaptive control law is proposed

$$\mathbf{f} = k_v \mathbf{s}_p + \alpha_p \tilde{\mathbf{p}} + \hat{m}(t) \boldsymbol{\psi}_p(\dot{\mathbf{v}}_r), \quad (12)$$

where k_v and α_p are positive constants and $\hat{m}(t)$ is an estimate of m which is updated according to the adaptive law

$$\dot{\hat{m}} = \gamma_p \boldsymbol{\psi}_p^T(\dot{\mathbf{v}}_r) \mathbf{s}_p, \quad (13)$$

where $\gamma_p \in \mathbb{R}^+$ is the adaptive gain.

3.2 | Computation of the desired orientation and total thrust

From Equations (1) and (10), it is clear that the quadrotor cannot follow an arbitrary desired trajectory with an arbitrary orientation. In fact, the quadrotor's position trajectory depends on its orientation. In order that the quadrotor tracks a desired trajectory the total thrust $\mathcal{T}$ and desired orientation $\mathbf{R}_d$ must be properly calculated. By exploiting the properties of rotation matrices, the total thrust can be computed as follows

$$\mathcal{T} = \|\mathbf{f}\| \quad (14)$$

with $\mathbf{f}$ given in (12). Now let us compute the desired orientation $\mathcal{Q}_d$. To this end, define the unit vector

$$\mathbf{r} = \frac{\mathbf{f}}{\|\mathbf{f}\|} = \text{col}(r_x, r_y, r_z) \quad (15)$$

and align the $\mathbf{z}_i$ -axis with $\mathbf{r}$ (see Figure 2). This can be done by the following rotation matrix

$$\mathbf{R}_a = \mathbf{a}\mathbf{a}^T + \cos(\phi)(\mathbf{I} - \mathbf{a}\mathbf{a}^T) + \sin(\phi)\mathbf{S}(\mathbf{a}), \quad (16)$$

where $\mathbf{I} \in \mathbb{R}^{3 \times 3}$ is the identity matrix, ϕ is the angle between $\mathbf{z}_i$ and $\mathbf{r}$ (see Figure 2), and $\mathbf{a}$ is the unit vector perpendicular to the plane formed by $\mathbf{z}_i$ and $\mathbf{r}$, that is,

$$\mathbf{a} = \frac{1}{\sin(\phi)} \mathbf{S}(\mathbf{z}_i) \mathbf{r} = \frac{1}{\sqrt{1 - r_z^2}} \begin{bmatrix} -r_y \\ r_x \\ 0 \end{bmatrix}. \quad (17)$$

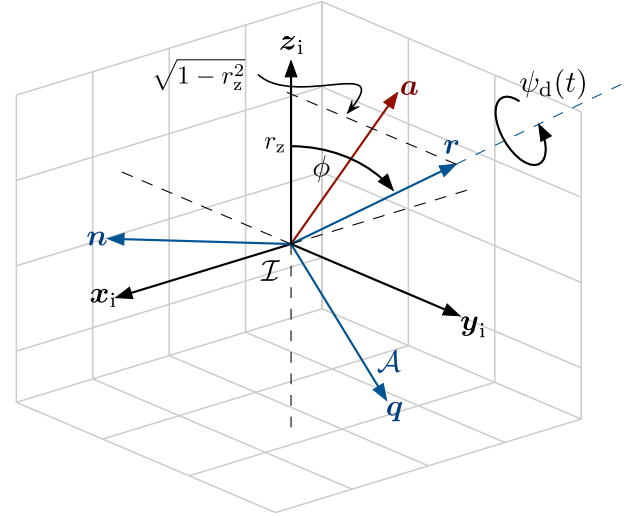
In terms of the unit quaternion one has

$$\mathcal{Q}_a = \{\cos(\phi/2), \sin(\phi/2)\mathbf{a}\}, \quad (18)$$

where

$$\cos(\phi/2) = \sqrt{\frac{1 + r_z}{2}}, \quad \sin(\phi/2) = \sqrt{\frac{1 - r_z}{2}}. \quad (19)$$

FIGURE 2 Computation of the desired rotation matrix [Colour figure can be viewed at wileyonlinelibrary.com]



The quaternion Q_a satisfies $Q_a \odot \mathbf{r} = \mathbf{r}$ and its time derivative is given $\dot{Q}_a = \frac{1}{2} Q_a \otimes \{0, \boldsymbol{\omega}_a\}$ where $\boldsymbol{\omega}_a \in \mathfrak{R}^3$ is the angular velocity expressed with respect to the frame $\mathcal{A} = \{\mathbf{n}, \mathbf{q}, \mathbf{r}\}$. A direct computation shows that the components of $\boldsymbol{\omega}_a$ are given by

$$\{0, \boldsymbol{\omega}_a\} = 2Q_a^{-1} \otimes \dot{Q}_a \Rightarrow \boldsymbol{\omega}_a = \frac{1}{1+r_z} \begin{bmatrix} -\dot{r}_y(1+r_z) + r_y\dot{r}_z \\ \dot{r}_x(1+r_z) - r_x\dot{r}_z \\ r_y\dot{r}_x - r_x\dot{r}_y \end{bmatrix}, \quad (20)$$

where $\dot{\mathbf{r}} = \|\mathbf{f}\|^{-1} (\mathbf{I} - \|\mathbf{f}\|^{-2} \mathbf{f} \mathbf{f}^T) \frac{d}{dt} \mathbf{f}$. At this stage, it is possible to set $Q_d = Q_a$ and $\boldsymbol{\omega}_d = \boldsymbol{\omega}_a$. However, the quadrotor can still rotate an arbitrary angle $\psi_d(t)$ around $\mathbf{r}$ without modifying the direction of the total force (15). Therefore, the desired orientation can be computed as

$$Q_d = Q_a \otimes Q_\psi, \quad (21)$$

where $Q_\psi = \{\cos(\psi_d(t)/2), \sin(\psi_d(t)/2)\mathbf{z}_i\}$. This leads to the desired angular velocity

$$\boldsymbol{\omega}_d(t) = \boldsymbol{\omega}_\psi(t) + Q_\psi^{-1} \odot \boldsymbol{\omega}_a, \quad (22)$$

where $\boldsymbol{\omega}_\psi = \text{col}(0, 0, \dot{\psi}_d(t)) \in \mathfrak{R}^3$.

The angle $\psi_d(t)$ plays a similar role of the yaw angle of the Euler angles parametrization. The proposed desired orientation gives an additional degree of freedom that allows to orient the quadrotor around the $\mathbf{r}$ -axis while is tracking a desired trajectory in a tridimensional space. Notice that (21) is equivalent to $\mathbf{R}_d = \mathbf{R}_a \mathbf{R}_\psi$ where $\mathbf{R}_\psi \in SO(3)$ is the basic rotation about the z -axis. It is important to point out that the desired unit quaternion (21) is valid whenever $\mathbf{r} \neq -\mathbf{z}_i$.

3.3 | Attitude adaptive control

Once the desired orientation is properly defined, the next step is to design the attitude controller. To this end, the orientation and angular velocity errors are defined as

$$\tilde{Q} = Q^{-1} \otimes Q_d = \{\tilde{\eta}, \tilde{\epsilon}\}, \quad \tilde{\boldsymbol{\omega}} = \tilde{Q} \odot \boldsymbol{\omega}_d(t) - \boldsymbol{\omega}. \quad (23)$$

Similar to the position case, an angular reference velocity $\boldsymbol{\omega}_r$ and auxiliary state variable $\mathbf{s}_0$ are introduced

$$\boldsymbol{\omega}_r = \tilde{Q} \odot \boldsymbol{\omega}_d(t) + k_\epsilon \tilde{\epsilon} + 2\alpha_0^{-1} \mathcal{T}\mathcal{S}(\tilde{\epsilon}) Q^{-1} \odot \mathbf{s}_p, \quad (24)$$

$$\mathbf{s}_0 = \boldsymbol{\omega}_r - \boldsymbol{\omega} = \tilde{\boldsymbol{\omega}} + k_\epsilon \tilde{\epsilon} + 2\alpha_0^{-1} \mathcal{T} \mathbf{S}(\bar{\epsilon}) \mathcal{Q}^{-1} \odot \mathbf{s}_p, \quad (25)$$

where $k_\epsilon, \alpha_0 \in \mathfrak{R}$ are positive constants, $\bar{\epsilon} = \text{col}(\tilde{\epsilon}_y, -\tilde{\epsilon}_x, \tilde{\eta}) \in \mathfrak{R}^3$, and $\mathbf{s}_p$ is defined in (9). In a similar fashion as the translation case, the attitude dynamics are written in terms of the orientation tracking errors $\tilde{\mathcal{Q}} = \{\tilde{\eta}, \tilde{\epsilon}\}$ and $\mathbf{s}_0$. By taking into account Equations (2), (5), and (23)–(25) the dynamics of the attitude errors are given by

$$\dot{\tilde{\mathcal{Q}}} = \mathbf{L}(\tilde{\eta}, \tilde{\epsilon}) \tilde{\boldsymbol{\omega}}, \quad (26a)$$

$$\tilde{\boldsymbol{\omega}} = -k_\epsilon \tilde{\epsilon} - 2\alpha_0^{-1} \mathcal{T} \mathbf{S}(\bar{\epsilon}) \mathcal{Q}^{-1} \odot \mathbf{s}_p + \mathbf{s}_0, \quad (26b)$$

$$\mathbf{J} \dot{\mathbf{s}}_0 = \boldsymbol{\Psi}_0(\boldsymbol{\omega}, \boldsymbol{\omega}_r, \dot{\boldsymbol{\omega}}_r) \boldsymbol{\theta} - \mathbf{S}(\mathbf{s}_0) \mathbf{J} \boldsymbol{\omega} - \boldsymbol{\tau}, \quad (26c)$$

where $\boldsymbol{\Psi}_0(\boldsymbol{\omega}, \boldsymbol{\omega}_r, \dot{\boldsymbol{\omega}}_r) \boldsymbol{\theta} = \mathbf{J} \dot{\boldsymbol{\omega}}_r + \mathbf{S}(\boldsymbol{\omega}_r) \mathbf{J} \boldsymbol{\omega}$. In this case, the control objective is to design $\boldsymbol{\tau}$ such that

$$\lim_{t \rightarrow \infty} \tilde{\mathcal{Q}} = \mathbf{1}_Q, \quad \lim_{t \rightarrow \infty} \tilde{\boldsymbol{\omega}} = \mathbf{0},$$

while assuming uncertainties in the inertia tensor $\mathbf{J}$. To fulfill this task, the following adaptive attitude control is proposed

$$\boldsymbol{\tau} = k_\omega \mathbf{s}_0 + \alpha_0 \tilde{\epsilon} + \boldsymbol{\Psi}_0(\boldsymbol{\omega}, \boldsymbol{\omega}_r, \dot{\boldsymbol{\omega}}_r) \hat{\boldsymbol{\theta}}(t), \quad (27)$$

where $k_\omega \in \mathfrak{R}$ is a positive constants and $\hat{\boldsymbol{\theta}}(t) \in \mathfrak{R}^p$ is updated by

$$\dot{\hat{\boldsymbol{\theta}}} = \gamma_0 \boldsymbol{\Psi}_0^T(\boldsymbol{\omega}, \boldsymbol{\omega}_r, \dot{\boldsymbol{\omega}}_r) \mathbf{s}_0, \quad (28)$$

where $\gamma_0 \in \mathfrak{R}$ is a positive constant.

3.4 | Stability analysis

The main contribution of the article is summarized in the following theorem:

Theorem 1. Assume that $\tilde{\eta}(0) \neq -1$ and consider the position and attitude dynamics described by (11) and (26) in closed-loop with the adaptive control laws (12)–(13) and (27)–(28). Then, the trajectories of the resulting closed-loop dynamics given by

$$\dot{\tilde{\mathbf{v}}} = -k_p \tilde{\mathbf{p}} + \mathbf{s}_p, \quad (29a)$$

$$m \dot{\mathbf{s}}_p = -k_v \mathbf{s}_p - \alpha_p \tilde{\mathbf{p}} - \tilde{m} \boldsymbol{\psi}_p(\dot{\mathbf{v}}_r) - \mathcal{T}(\mathbf{R} - \mathbf{R}_d) \mathbf{z}_i, \quad (29b)$$

$$\dot{\tilde{m}} = \gamma_p \boldsymbol{\psi}_p^T(\dot{\mathbf{v}}_p) \mathbf{s}_p, \quad (29c)$$

where $\tilde{m} = \hat{m}(t) - m$ and

$$\dot{\tilde{\mathcal{Q}}} = \mathbf{L}(\tilde{\eta}, \tilde{\epsilon}) \tilde{\boldsymbol{\omega}}, \quad (30a)$$

$$\tilde{\boldsymbol{\omega}} = -k_\epsilon \tilde{\epsilon} - 2\alpha_0^{-1} \mathcal{T} \mathbf{S}(\bar{\epsilon}) \mathcal{Q}^{-1} \odot \mathbf{s}_p + \mathbf{s}_0, \quad (30b)$$

$$\mathbf{J} \dot{\mathbf{s}}_0 = -\mathbf{S}(\mathbf{s}_0) \mathbf{J} \boldsymbol{\omega} - k_\omega \mathbf{s}_0 - \alpha_0 \tilde{\epsilon} - \boldsymbol{\Psi}_0(\boldsymbol{\omega}, \boldsymbol{\omega}_r, \dot{\boldsymbol{\omega}}_r) \tilde{\boldsymbol{\theta}}, \quad (30c)$$

$$\dot{\tilde{\boldsymbol{\theta}}} = \gamma_0 \boldsymbol{\Psi}_0^T(\boldsymbol{\omega}, \boldsymbol{\omega}_r, \dot{\boldsymbol{\omega}}_r) \mathbf{s}_0, \quad (30d)$$

with $\tilde{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}$, are bounded for all $t \geq 0$. Moreover, the position and attitude tracking errors converge asymptotically to zero, that is,

$$\lim_{t \rightarrow \infty} \tilde{\mathbf{p}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\mathbf{v}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\epsilon} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\omega} = \mathbf{0} \quad (31)$$

and $\tilde{\eta} \rightarrow 1$ as $t \rightarrow \infty$.

Proof. First, let us analyze the interconnection term $(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i$, which can be expressed as

$$(\mathbf{R} - \mathbf{R}_d)\mathbf{z}_i = \mathbf{R}(\mathbf{z}_i - \tilde{\mathbf{R}}\mathbf{z}_i) = \mathbf{Q} \odot (\mathbf{z}_i - \tilde{\mathbf{Q}} \odot \mathbf{z}_i), \quad (32)$$

where $\tilde{\mathbf{R}} = \mathbf{R}^T \mathbf{R}_d$ is the rotation matrix error. Furthermore, a simple calculation shows that the term $\mathbf{z}_i - \tilde{\mathbf{Q}} \odot \mathbf{z}_i$ can be written as

$$\mathbf{z}_i - \tilde{\mathbf{Q}} \odot \mathbf{z}_i = 2 \begin{bmatrix} -\tilde{\epsilon}_x \tilde{\epsilon}_z - \tilde{\eta} \tilde{\epsilon}_y \\ -\tilde{\epsilon}_y \tilde{\epsilon}_z + \tilde{\eta} \tilde{\epsilon}_x \\ \tilde{\epsilon}_x^2 + \tilde{\epsilon}_y^2 \end{bmatrix} = 2\mathbf{S}(\bar{\epsilon})\bar{\epsilon}, \quad (33)$$

where $\bar{\epsilon} = \text{col}(\tilde{\epsilon}_y, -\tilde{\epsilon}_x, \tilde{\eta})$ and $\tilde{\eta}^2 + \bar{\epsilon}^T \bar{\epsilon} = 1$ has been used.

Now, consider the candidate Lyapunov function

$$V = \frac{1}{2} \alpha_p \|\tilde{\mathbf{p}}\|^2 + \frac{1}{2} m \|\mathbf{s}_p\|^2 + \frac{1}{2} \gamma_p^{-1} \tilde{m}^2 + \alpha_o \|\tilde{\epsilon}\|^2 + \alpha_o (1 - \tilde{\eta})^2 + \frac{1}{2} \mathbf{s}_o^T \mathbf{J} \mathbf{s}_o + \frac{1}{2} \gamma_o^{-1} \|\tilde{\theta}\|^2, \quad (34)$$

whose time derivative along (29) and (30) is given by

$$\begin{aligned} \dot{V} &= -\alpha_p k_p \|\tilde{\mathbf{p}}\|^2 - k_v \|\mathbf{s}_p\|^2 - 2\mathcal{T} \mathbf{s}_p^T \mathbf{Q} \odot \mathbf{S}(\bar{\epsilon})\bar{\epsilon} + \alpha_o \tilde{\epsilon}^T \tilde{\omega} - k_\omega \|\mathbf{s}_o\|^2 - \alpha_o \mathbf{s}_o^T \tilde{\epsilon} \\ &= -\alpha_p k_p \|\tilde{\mathbf{p}}\|^2 - \alpha_o k_\epsilon \|\tilde{\epsilon}\|^2 - k_v \|\mathbf{s}_p\|^2 - k_\omega \|\mathbf{s}_o\|^2 - 2\mathcal{T} (\mathbf{s}_p^T \mathbf{Q} \odot \mathbf{S}(\bar{\epsilon})\bar{\epsilon} + \tilde{\epsilon}^T \mathbf{S}(\bar{\epsilon})\mathbf{Q}^{-1} \odot \mathbf{s}_p). \end{aligned} \quad (35)$$

By taking into account that $\tilde{\epsilon}^T \mathbf{S}(\bar{\epsilon})\mathbf{Q}^{-1} \odot \mathbf{s}_p = \mathbf{s}_p^T \mathbf{R} \mathbf{S}^T(\bar{\epsilon})\tilde{\epsilon} = -\mathbf{s}_p^T \mathbf{Q} \odot \mathbf{S}(\bar{\epsilon})\tilde{\epsilon}$, one gets

$$\dot{V} = -\alpha_p k_p \|\tilde{\mathbf{p}}\|^2 - \alpha_o k_\epsilon \|\tilde{\epsilon}\|^2 - k_v \|\mathbf{s}_p\|^2 - k_\omega \|\mathbf{s}_o\|^2 \leq 0. \quad (36)$$

Since the time derivative of V is negative semidefinite, the position and attitude tracking errors $\tilde{\mathbf{p}}$, $\tilde{\epsilon}$, $\tilde{\eta}$, $\mathbf{s}_p$, $\mathbf{s}_o$ and the parametric errors $\tilde{m}$, $\tilde{\theta}$ are bounded. This implies that $\tilde{\mathbf{v}}$, $\mathbf{p}$, $\mathbf{v}$, $\hat{m}$, and $\hat{\theta}$ are also bounded. From (12) and (14), the total thrust can be upper bounded as follows

$$\mathcal{T} \leq \alpha_o \|\tilde{\mathbf{p}}\| + k_v \|\mathbf{s}_p\| + |\hat{m}| (\|\mathbf{v}_d\|_\infty + k_p \|\tilde{\mathbf{v}}\| + g), \quad (37)$$

where the property $\|\mathbf{R}_d\| = 1$ has been used. Therefore, from (30) and (37) it is concluded that $\tilde{\omega}$ and ω are bounded vectors. Define the extended state $\mathbf{x} = \text{col}(\tilde{\mathbf{p}}, \mathbf{s}_p, \tilde{m}, \tilde{\epsilon}, \tilde{\eta}, \mathbf{s}_o, \tilde{\theta}) \in \mathfrak{R}^{14+p}$ with $\tilde{\eta} = 1 - \tilde{\eta}$. Then, the Lyapunov function (34) becomes

$$V = \frac{1}{2} \mathbf{x}^T \mathbf{P} \mathbf{x}, \quad (38)$$

where $\mathbf{P} = \text{blockdiag}\{\mathbf{P}_p, \mathbf{P}_o\} \in \mathfrak{R}^{14+p \times 14+p}$ is a block diagonal positive definite matrix with

$$\mathbf{P}_p = \begin{bmatrix} \alpha_p \mathbf{I} & \mathbf{O} & \mathbf{0} \\ \mathbf{O} & m \mathbf{I} & \mathbf{0} \\ \mathbf{0}^T & \mathbf{0}^T & \gamma_p^{-1} \end{bmatrix}, \quad \mathbf{P}_o = \begin{bmatrix} 2\alpha_o \mathbf{I} & \mathbf{O} & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & 2\alpha_o \mathbf{I} & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{J} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{O} & \gamma_o^{-1} \mathbf{I} \end{bmatrix}, \quad (39)$$

where $\mathbf{O}$ denotes the zero matrix with proper dimension. The function V satisfies

$$\frac{1}{2} \lambda_{\min}\{\mathbf{P}\} \|\mathbf{x}\|^2 \leq V(\mathbf{x}) \leq \frac{1}{2} \lambda_{\max}\{\mathbf{P}\} \|\mathbf{x}\|^2, \quad (40)$$

where $\lambda_{\min}\{\cdot\}$ and $\lambda_{\max}\{\cdot\}$ denote the minimum and maximum eigenvalue of a matrix, respectively. Clearly, the closed-loop dynamics (29)–(30) and the function (38) satisfy the conditions of Theorem 2 (see Appendix) with

$$W_1(\mathbf{x}) = \frac{1}{2} \lambda_{\min}\{\mathbf{P}\} \|\mathbf{x}\|^2, \quad W_2(\mathbf{x}) = \frac{1}{2} \lambda_{\max}\{\mathbf{P}\} \|\mathbf{x}\|^2, \quad W_3(\mathbf{x}) = -\mathbf{x}^T \mathbf{Q} \mathbf{x}, \quad (41)$$

where $\mathbf{Q} = \text{blockdiag}\{\alpha_p k_p \mathbf{I}, k_v \mathbf{I}, 0, \alpha_o k_e \mathbf{I}, 0, k_\omega \mathbf{I}, \mathbf{O}\}$ is block diagonal positive semidefinite matrix. Then, it follows

$$\lim_{t \rightarrow \infty} \tilde{\mathbf{p}} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \mathbf{s}_p = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \tilde{\epsilon} = \mathbf{0}, \quad \lim_{t \rightarrow \infty} \mathbf{s}_o = \mathbf{0}. \quad (42)$$

The foregoing result also implies that the velocity errors $\tilde{\mathbf{v}}$ and $\tilde{\boldsymbol{\omega}}$ converge asymptotically to zero (see Equations 29a and 30a). Recalling that $\tilde{\eta}^2 + \tilde{\epsilon}^T \tilde{\epsilon} = 1$, one has $\tilde{\eta} \rightarrow \pm 1$ as $t \rightarrow \infty$. The control law (27) generates two closed-loop equilibrium points $\mathbf{1}_Q$ (stable) and $-\mathbf{1}_Q$ (unstable).^{34,35} Since it is assumed that $\tilde{\eta}(0) \neq -1$, the closed-loop trajectory moves to the stable equilibrium point, that is, $\tilde{\eta} \rightarrow 1$ as $t \rightarrow \infty$. This concludes the proof. ■

Remark 1. The adaptive control laws (10) and (27) can be written as

$$\mathbf{f} = k_v \tilde{\mathbf{v}} + (k_v k_p + \alpha_p) \tilde{\mathbf{p}} + \hat{m}(t) \boldsymbol{\psi}_p, \quad (43)$$

$$\boldsymbol{\tau} = k_\omega \tilde{\boldsymbol{\omega}} + (k_\omega k_e + \alpha_o) \tilde{\epsilon} + 2k_\omega \alpha_o^{-1} \mathcal{T} \mathbf{S}(\tilde{\epsilon}) \tilde{\epsilon} \mathbf{Q}^{-1} \odot \mathbf{s}_p + \boldsymbol{\Psi}_o \hat{\boldsymbol{\theta}}(t). \quad (44)$$

Therefore, the control scheme can be viewed as PD controller plus a nonlinear term. The control gains k_v , k_p , and k_ω , k_e can be tuned as PD gains. On the other hand, the gains α_p and α_o improve the convergence rate of the position and attitude tracking errors (see (36)) and they should be tuned accordingly to the desired convergence rate and transient response. In general, we can set $k_p > \alpha_p$ and $k_e > \alpha_o$. Finally, the adaptive gains γ_p and γ_o must be set small enough and slightly increase them until the desired performance is achieved.

4 | SIMULATION RESULTS

In this section, a set of numerical simulations are presented to show the performance of the adaptive controller. The dynamic parameters of the quadrotor used in the simulations were $m = 1$ kg, $\mathbf{J} = \text{diag}\{\theta_1, \theta_2, \theta_3\} = \{0.075, 0.075, 0.015\}$ kg m². The initial position and velocity of the aerial robot were chosen as $\mathbf{p}(0) = \text{col}(4.0, 0.5, 1.0)$ m, and $\mathbf{v}(0) = \text{col}(0, 0, -0.2)$ m/s, while its attitude initial configuration was chosen as $\boldsymbol{\omega}(0) = \text{col}(0.5, 0, 0.5)$ rad/s and $\mathbf{Q}(0) = \{\sqrt{2}/2, 0, 0, \sqrt{2}/2\}$ which correspond to the initial orientation

$$\mathbf{R}(0) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The reference trajectories are given by

$$\mathbf{p}_d(t) = \begin{bmatrix} 3 \cos\left(\frac{\pi}{20}t\right) \\ 3 \sin\left(\frac{\pi}{20}t\right) \\ \frac{1}{2} \cos\left(\frac{\pi}{5}t\right) + 2 \end{bmatrix} \text{ m}, \quad \psi_d(t) = \frac{\pi}{4} \sin\left(\frac{\pi}{10}t\right) \text{ rad}.$$

To avoid complex calculations the time derivatives of $\mathbf{f}$ and $\boldsymbol{\omega}_r$ were approximated by a first-order filter of the form $H(s) = s/(\epsilon s + 1)$ with $\epsilon = 0.05$. In order to resemble a more realistic scenario white Gaussian noise is added to the position/attitude and velocity signals. The control and adaptive gains used in the simulations are shown in Table 1. The performance of the proposed controller is compared with the nonlinear geometric control proposed by Lee et al.¹² given by

$$\boldsymbol{\tau} = (k_x \tilde{\mathbf{p}} + k_b \tilde{\mathbf{v}} + \bar{m}(\dot{\mathbf{v}}_d(t) + \mathbf{g} \mathbf{z}_i))^T \mathbf{R} \mathbf{z}_i, \quad (45)$$

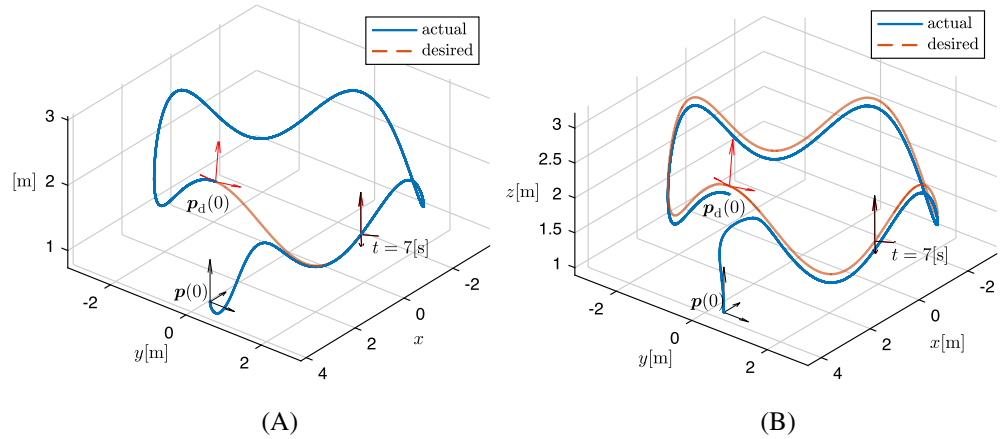
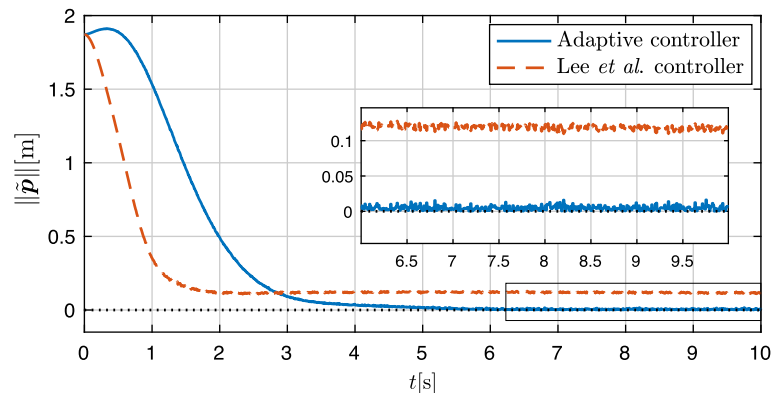
TABLE 1 Control and adaptive gains used in the simulation

Position				Attitude			
Adaptive controller		Lee et al. controller		Adaptive controller		Lee et al. controller	
Gain	Value	Gain	Value	Gain	Value	Gain	Value
k_p	1.25	k_x	10	k_ϵ	6.5	k_R	10
k_v	1	k_b	5	k_ω	3.5	k_Ω	4
α_p	1			α_o	2		
γ_p	0.03			γ_o	0.001		

$$\tau = k_R \mathbf{e}_R + k_\Omega \tilde{\omega} + \mathbf{S}(\omega) \mathbf{J} \omega + \mathbf{J} (\tilde{\mathbf{R}} \dot{\omega}_d - \mathbf{S}(\omega) \tilde{\mathbf{R}} \omega_d), \quad (46)$$

where $\bar{m} \in \mathfrak{R}$ is the nominal mass, k_x, k_b, k_R , and $k_\Omega \in \mathfrak{R}$ are positive control gains and the attitude error $\mathbf{e}_R$ is defined as $\mathbf{e}_R = \text{vex} \left(\frac{1}{2} \tilde{\mathbf{R}} - \frac{1}{2} \tilde{\mathbf{R}}^T \right)$ where $\text{vex}(\cdot) : \mathfrak{so}(3) \rightarrow \mathfrak{R}^3$ is the inverse of the operator $\mathbf{S}(\cdot)$. In the absence of parametric uncertainties, the previous controller guarantees almost globally exponentially tracking of the references trajectories. In the simulations, only uncertainty in the mass of the quadrotor is considered. The nominal mass was set as $\bar{m} = 0.9m = 0.9 \text{ kg}$ (10% of discrepancy). The control gains of the Lee's controller used in the simulations are also depicted in Table 1. For the adaptive control law, the initial values of $\hat{m}$ and $\hat{\theta}$ was chosen as $\hat{m}(0) = 0$ and $\hat{\theta} = \mathbf{0}$, respectively.

The trajectory of the quadrotor in the tridimensional space is shown in Figure 3. This figure also shows the actual and desired orientation at the time instants $t = 0 \text{ s}$ and $t = 7 \text{ s}$. Despite parametric uncertainties, the proposed adaptive controller successfully achieves trajectory tracking of the reference position. On the other hand, the mass uncertainty causes a steady state error for the Lee et al. controller. Figure 4 shows the norm of the position errors. Due to the added noise, the tracking errors converge nearly to zero. In this figure it can be appreciated that the position errors obtained with

**FIGURE 3** Trajectory of the quadrotor in the 3D environment: (A) adaptive controller and (B) Lee et al. controller [Colour figure can be viewed at wileyonlinelibrary.com]**FIGURE 4** Norm of the position tracking errors [Colour figure can be viewed at wileyonlinelibrary.com]

the Lee et al. controller converge faster to zero than the ones obtained with the proposed adaptive controller. Nevertheless, due to parametric uncertainties a steady state error is present after the transient response. As it is expected, the proposed controller compensates such uncertainties and the tracking errors converge asymptotically to zero.

The attitude controller (46) is directly designed on $SO(3)$. To make a comparison of the attitude performance the unit quaternion is extracted from the rotation matrix $\tilde{\mathbf{R}} = \mathbf{R}^T \mathbf{R}_d$ as follows³⁶

$$\tilde{\eta} = \frac{1}{2} \sqrt{\text{tr}(\tilde{\mathbf{R}}) + 1}, \quad \tilde{\epsilon} = \frac{1}{4\tilde{\eta}} \begin{bmatrix} \tilde{r}_{32} - \tilde{r}_{23} \\ \tilde{r}_{13} - \tilde{r}_{31} \\ \tilde{r}_{21} - \tilde{r}_{12} \end{bmatrix},$$

where $\text{tr}(\cdot)$ is trace operator and $\tilde{r}_{ij}$ is the ij th element of $\tilde{\mathbf{R}}$. The attitude errors are shown in Figure 5. In this case, the behavior obtained with each controller is quite similar. The estimated mass and inertia parameters are shown in Figure 6. The estimated mass converge to the actual value, this implies that the translational regressor $\boldsymbol{\psi}_p(\dot{\mathbf{v}}_r)$ satisfies the persistency excitation condition³⁷ (PE),

$$\int_t^{t+T} \boldsymbol{\psi}_p^T(\dot{\mathbf{v}}_r(\vartheta)) \boldsymbol{\psi}_p(\dot{\mathbf{v}}_r(\vartheta)) d\vartheta \geq \delta, \quad T > 0, \delta > 0.$$

On the other, the estimated inertia parameters does not converge to their true values. The reference trajectory was not a sufficiently “rich” signal to guarantee that the attitude regressor $\boldsymbol{\Psi}_o(\boldsymbol{\omega}, \boldsymbol{\omega}_r, \dot{\boldsymbol{\omega}}_r)$ is persistently exciting. This is not a drawback since the objective is trajectory tracking rather than parameter estimation. Finally, the total thrust and the input torque are shown in Figure 7. To compensate the mass uncertainty a high gain was used in the simulations for the Lee’s position controller. This leads to a thrust overshoot during the transient phase as it can be appreciated in Figure 7(A). Since the adaptive controller must compensate not only the initial errors but also the model uncertainties, more control effort is required for the attitude subsystem (see Figure 7(B)).

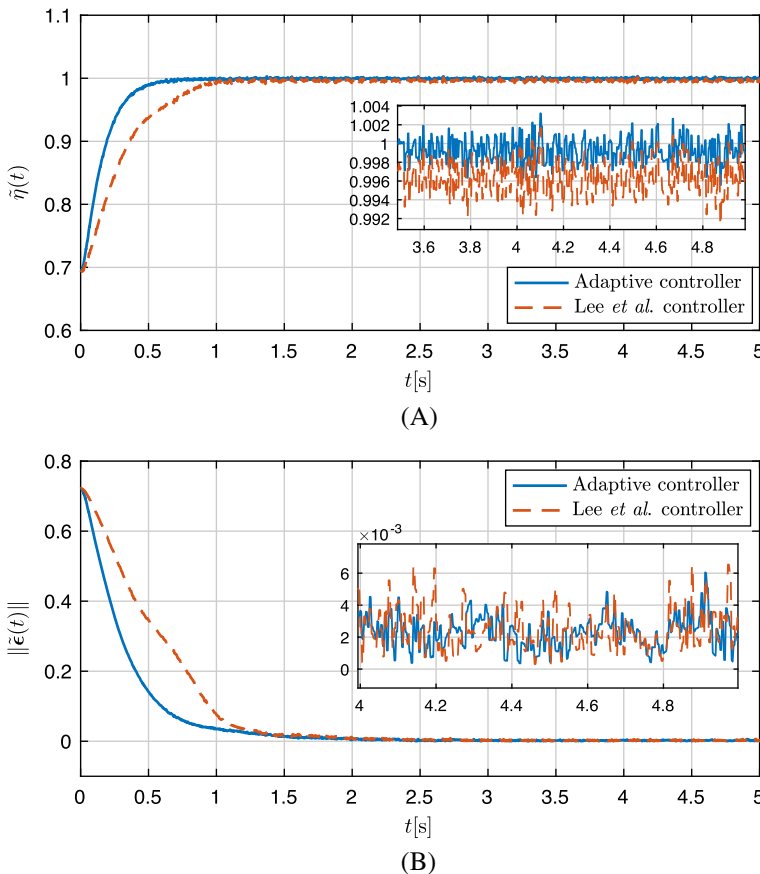


FIGURE 5 Attitude errors: (A) $\tilde{\eta}$ and (B) $\|\tilde{\epsilon}\|$
[Colour figure can be viewed at wileyonlinelibrary.com]

FIGURE 6 Trajectory of the estimated parameters: (A) mass and (B) inertia parameters [Colour figure can be viewed at wileyonlinelibrary.com]

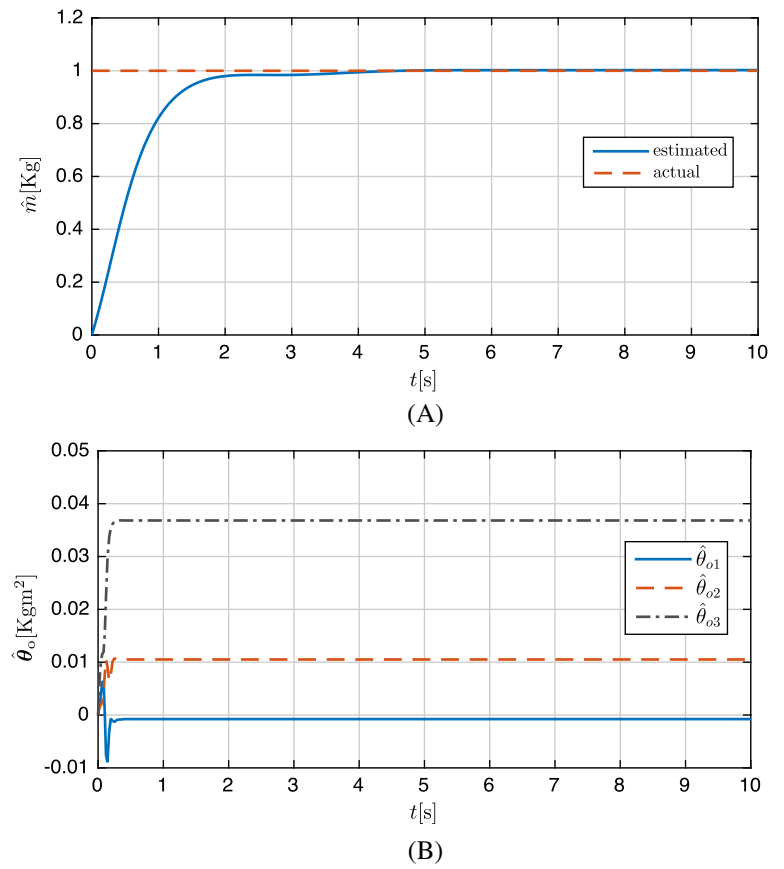
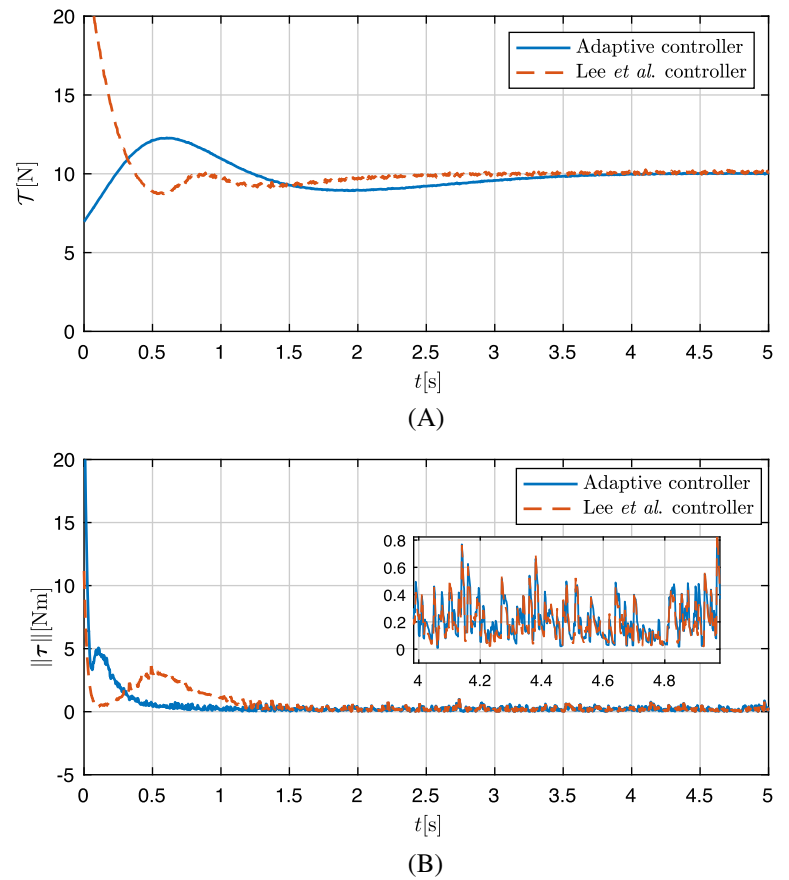


FIGURE 7 Control inputs: (A) total thrust $\mathcal{T}$ and (B) torque $\|\tau\|$ [Colour figure can be viewed at wileyonlinelibrary.com]



5 | CONCLUSIONS

In this work, the problem of trajectory tracking of a quadrotor with mass and inertia uncertainties is studied. To tackle model uncertainties, a hierarchical adaptive control based on the unit quaternion is proposed which guarantees that the position and attitude tracking errors converge asymptotically to zero. The attitude control takes into account information of the translational errors. Therefore, the cross terms that appear during the stability analysis are properly canceled. The stability analysis of the complete closed-loop system is carried out based on Lyapunov theory and does not rely on the properties of cascade systems. The performance of the proposed controller was validated by means of numerical simulations. The outcomes show a good performance even with noise measurements. Experimental validation and designing robust adaptive controllers will be considered as a future research.

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DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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APPENDIX A

Consider following auxiliary theorem³⁸

Theorem 2. Consider the system $\dot{\mathbf{x}} = \mathbf{g}(t, \mathbf{x})$. Let D be a domain containing $\mathbf{x} = \mathbf{0}$ and suppose that $\mathbf{g}(t, \mathbf{x})$ is continuous in t and locally Lipchitz in $\mathbf{x}$, uniformly in t on $[0, \infty) \times D$. Furthermore, suppose that $\mathbf{g}(t, \mathbf{0})$ is uniformly bounded for all $t \geq 0$. Let $V : [0, \infty) \times D \rightarrow \mathbb{R}$ be a continuously differentiable function such that

$$W_1(\mathbf{x}) \leq V(t, \mathbf{x}) \leq W_2(\mathbf{x}),$$

$$\dot{V}(t, \mathbf{x}) = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial \mathbf{x}} \mathbf{g}(t, \mathbf{x}) \leq -W_3(\mathbf{x})$$

for all $t \geq 0$, for all $\mathbf{x} \in D$, where $W_1(\mathbf{x})$ and $W_2(\mathbf{x})$ are continuous positive definite functions and $W_3(\mathbf{x})$ is a continuous positive semidefinite function on D . Choose $r > 0$ such that $B_d \subset D$ and let $\rho < \min_{\|\mathbf{x}\|=r} W_1(\mathbf{x})$. Then, all the solutions of $\dot{\mathbf{x}} = \mathbf{g}(t, \mathbf{x})$ are bounded and satisfy

$$W_3(\mathbf{x}) \rightarrow 0 \quad \text{as} \quad t \rightarrow \infty.$$

Moreover, if all assumptions hold globally and $W_1(\mathbf{x})$ is radially unbounded, the statement is true for all $\mathbf{x}(t_0) \in \mathbb{R}^n$.